$$\sigma^{-1}[(a_1,a_2,a_3)\sigma(a_1,a_2,a_3)^{-1}] = (a_1,a_2)(a_3,a_4)\mu^{-1}(a_1,a_2,a_3)\mu(a_3,a_4)(a_1,a_2)(a_1,a_3,a_2) = (a_1,a_3)(a_2,a_4)$$

is in N. Continuing with the hint given, we let $\alpha = (a_1,a_3)(a_2,a_4)$ and $\beta = (a_1,a_3,i)$ where i is different from a_1, a_2, a_3, and a_4. Then $\beta \in A_n$ and $\alpha \in N$ and N a normal subgroup of A_n imply that $(\beta^{-1}\alpha\beta)\alpha \in N$. Computing, we find that

$$(\beta^{-1}\alpha\beta)\alpha = (a_1,i,a_3)(a_1,a_3)(a_2,a_4)(a_1,a_3,i)(a_1,a_3)(a_2,a_4) = (a_1,a_3,i).$$

Thus $N = A_n$ in this case also, by Part(**d**).

40. *Closure:* Let $h_1n_1, h_2n_2 \in HN$ where $h_1, h_2 \in H$ and $n_1, n_2 \in N$. Because N is a normal subgroup, left cosets are right cosets so $Nh_2 = h_2N$; in particular, $n_1h_2 = h_2n_3$ for some $n_3 \in N$. Then $(h_1n_1)(h_2n_2) = (h_1h_2)(n_3n_2) \in HN$, so HN is closed under the group operation.

Identity: Because $e \in H$ and $e \in N$, we see that $e = ee \in HN$.

Inverses: Now $(h_1n_1)^{-1} = n_1^{-1}h_1^{-1} \in Nh_1^{-1}$ and $Nh_1^{-1} = h_1^{-1}N$ because N is normal. Thus $n_1^{-1}h_1^{-1} = h_1^{-1}n_4$ for some $n_4 \in N$, so $(h_1n_1)^{-1} \in HN$, and we see that HN is a subgroup of G.

Clearly HN is the smallest subgroup of G containing both H and N, because any such subgroup must contain all the products hn for $h \in H$ and $n \in N$.

41. Exercise 40 shows that NM is a subgroup of G. We must show that $g(nm)g^{-1} \in NM$ for all $g \in G, n \in N$, and $m \in M$. We have $g(nm)g^{-1} = (gng^{-1})(gmg^{-1})$. Because N and M are both normal, we know that $gng^{-1} \in N$ and $gmg^{-1} \in M$. Thus $g(nm)g^{-1} \in NM$ so NM is a normal subgroup of G.

42. The fact that K is normal shows that $hkh^{-1} \in K$, so $(hkh^{-1})k^{-1} \in K$. The fact that H is normal shows that $kh^{-1}k^{-1} \in H$, so $h(kh^{-1}k^{-1}) \in H$. Thus $hkh^{-1}k^{-1} \in H \cap K$, so $hkh^{-1}k^{-1} = e$. It follows that $hk = kh$.

16. Group Action on a Set

1. (See the answer in the text.)

2. $G_1 = G_3 = \{\rho_0, \delta_2\}$, $G_2 = G_4 = \{\rho_0, \delta_1\}$,
$G_{s_1} = G_{s_3} = \{\rho_0, \mu_1\}$, $G_{s_2} = G_{s_4} = \{\rho_0, \mu_2\}$,
$G_{m_1} = G_{m_2} = \{\rho_0, \rho_2, \mu_1, \mu_2\}$, $G_{d_1} = G_{d_2} = \{\rho_0, \rho_2, \delta_1, \delta_2\}$,
$G_C = G$, $G_{P_1} = G_{P_3} = \{\rho_0, \mu_1\}$, $G_{P_2} = G_{P_4} = \{\rho_0, \mu_2\}$

3. $\{1,2,3,4\}$, $\{s_1, s_2, s_3, s_4\}$, $\{m_1, m_2\}$, $\{d_1, d_2\}$, $\{C\}$, $\{P_1, P_2, P_3, P_4\}$

4. The definition is incorrect. We need a universal quantifier.

A group G **acts faithfully** on X if and only if $gx = x$ for all $x \in G$ implies that $g = e$.

5. The definition is incorrect, and is an example of a nonsense definition.

A group G is **transitive** on a G-set X if and only if for each $a, b \in X$, there exists some $g \in G$ such that $ga = b$.

6. Every sub-G-set of a G-set X consists of a union of orbits in X under G.

7. A G-set if transitive if and only if it has only one orbit.

8. F T F T F T T F T T

9. **a.** $\{P_1, P_2, P_3, P_4\}$ and $\{s_1, s_2, s_3, s_4\}$ are isomorphic sub-D_4-sets. Note that if you change each P to an s in Table 16.10, you get a duplication of the four columns for s_1, s_2, s_3, and s_4.

b. δ_1 leaves two elements, 2 and 4, of $\{1, 2, 3, 4\}$ fixed, but δ_1 leaves no elements of $\{s_1, s_2, s_3, s_4\}$ fixed.

c. Yes, for after Part(**b**), the only other conceivable choice for an isomorphism is $\{m_1, m_2\}$ with $\{d_1, d_2\}$. However, μ_1 leaves the elements of $\{m_1, m_2\}$ fixed and moves both elements of $\{d_1, d_2\}$ so they are not isomorphic.

10. **a.** Yes, for ρ_0 is the only element of G that leaves every element of X fixed.

b. $\{1, 2, 3, 4\}$, $\{s_1, s_2, s_3, s_4\}$, *and* $\{P_1, P_2, P_3, P_4\}$

11. Let $g_1, g_2 \in G$. Now suppose that $g_1a = g_2a$ for all $a \in X$, which is true if and only if and only if ${g_2}^{-1}g_1a = a$ for all $a \in X$. If $g_1 \neq g_2$, then ${g_2}^{-1}g_1 \neq e$, and the action of G on X is not faithful. If the action of G on X is faithful, then we must have ${g_2}^{-1}g_1 = e$ and $g_1 = g_2$, that is, two distinct elements of G cannot act the same on each $a \in G$.

12. *Closure:* Let $g_1, g_2 \in G_Y$. Then for each $y \in Y$, we have $(g_1g_2)y = g_1(g_2y) = g_1y = y$, so $g_1g_2 \in G_Y$, and G_Y is closed under the group operation.

Identity: Because $ey = y$ for all $y \in Y$, we see that $e \in G_Y$.

Inverses: From $y = g_1y$ for all $y \in Y$, it follows that ${g_1}^{-1}y = {g_1}^{-1}(g_1y) = ({g_1}^{-1}g_1)y = ey = y$ for all $y \in Y$, so ${g_1}^{-1} \in G_Y$ also, and consequently $G_Y \leq G$.

13. **a.** Because rotation through 0 radians leaves each point of the plane fixed, the first requirement of Definition 16.1 is satisfied. The second requirement $(\theta_1 + \theta_2)P = \theta_1(\theta_2)P$ is also valid, because a rotation counterclockwise through $\theta_1 + \theta_2$ radians can be achieved by sequentially rotating through θ_2 radians and then through θ_1 radians.

b. The orbit containing P is a circle with center at the origin (0, 0) and radius the distance from P to the origin.

c. The group G_P is the cyclic subgroup $\langle 2\pi \rangle$ of G.

14. **a.** Let $X = \bigcup_{i \in I} X_i$ and let $x \in X$. Then $x \in X_i$ for precisely one index $i \in I$ because the sets are disjoint, and we define gx for each $g \in G$ to be the value given by the action of G on X_i. Conditions (1) and (2) in Definition 16.1 are satisfied because X_i is a G-set by assumption.

b. We have seen that each orbit in X is a sub-G-set. The G-set X can be regarded as the union of these sub-G-sets because the action gx of $g \in G$ on $x \in X$ coincides with the sub-G-set action gx of $g \in G$ on the same element x viewed as an element of its orbit.

15. Let $\phi : X \to L$ be defined by $\phi(x) = gG_{x_0}$ where $gx_0 = x$. Because G is transitive on X, we know that such a g exists. We must show that ϕ is well defined. Suppose that $g_1x_0 = x$ and $g_2x_0 = x$. Then $g_2x_0 = g_1x_0$ so $({g_1}^{-1}g_2)x_0 = x_0$. But then ${g_1}^{-1}g_2 \in G_{x_0}$ so $g_2 \in g_1G_{x_0}$ and $g_1G_{x_0} = g_2G_{x_0}$. This shows that the definition of $\phi(x)$ is independent of the choice of g such that $gx = x_0$, that is, ϕ is well defined.

It remains to show that ϕ is one to one and onto L, and that $g\phi(x) = \phi(gx)$ for all $x \in X$ and $g \in G$. Suppose that $\phi(x_1) = \phi(x_2)$ for $x_1, x_2 \in X$, and let $g_1x_0 = x_1$ and $g_2x_0 = x_2$. Then $\phi(x_1) = \phi(x_2)$ implies that $g_1G_{x_0} = g_2G_{x_0}$ so $g_2 = g_1g_0$ for some $g_0 \in G_{x_0}$. The equation $g_2x_0 = x_2$ then yields $g_1g_0x_0 = x_2$. Because $g_0 \in G_{x_0}$, we then obtain $g_1x_0 = x_2$ so $x_1 = x_2$ and ϕ is one to one. If $g \in G$, then $\phi(gx_0) = gG_{x_0}$ shows that ϕ maps X onto L. Finally, to show that $g\phi(x) = \phi(gx)$, let $x = g_1x_0$. Then $gx = g(g_1x_0) = (gg_1)x_0$ so $\phi(gx) = (gg_1)G_{x_0} = g(g_1G_{x_0}) = g\phi(x)$.

16. By Exercise 14, each G-set X is the union of its G-set orbits X_i for $i \in I$. By Exercise 15, each G-set orbit X_i is isomorphic to a G-set consisting of left cosets of $G_{x_{i,0}}$ where $x_{i,0}$ is any point of X_i. It is possible that the group $G_{x_{i,0}}$ may the the same as the group $G_{x_{j,0}}$ for some $j \neq i$ in I, but by attaching the index i to each coset of $G_{x_{i,0}}$ and j to each coset of $G_{x_{j,0}}$ as indicated in the statement of the exercise, we can consider these ith and jth coset G-sets to be disjoint. Identifying X_i with this isomorphic ith coset G-set, we see that X is isomorphic to a disjoint union of left coset G-sets.

17. a. If $g \in K$ so that $g(g_0x_0) = g_0x_0$, then $(g_0^{-1}gg_0)x_0 = x_0$, which means that $g_0^{-1}gg_0 \in H$, so $g \in g_0Hg_0^{-1}$. Because g may be any element of K, this shows that $K \subseteq g_0Hg_0^{-1}$. Making a symmetric argument, starting with $g \in H$, g_0x_0 as initial base point, and obtaining x_0 as second base point by g_0^{-1} acting on g_0x_0, we see that $H \subseteq g_0^{-1}Kg_0$, or equivalently, $g_0Hg_0^{-1} \subseteq K$. Thus $K = g_0Hg_0^{-1}$.

b. *Conjecture:* The G-set of left cosets of H is isomorphic to the G-set of left cosets of K if and only if H and K are conjugate subgroups of G.

c. We first show that if H and K are conjugate subgroups of G, then the G-set L_H of left cosets of H is isomorphic to the G-set L_K of left cosets of K. Let $g_0 \in G$ be chosen such that $K = g_0Hg_0^{-1}$. Note that for $aH \in L_H$, we have $aHg_0^{-1} = ag_0^{-1}g_0Hg_0^{-1} = ag_0^{-1}K \in L_K$. We define $\phi : L_H \to L_K$ by $\phi(aH) = ag_0^{-1}K$. We just saw that $ag_0^{-1}K = (aH)g_0^{-1}$ so ϕ is independent of the choice of $a \in H$, that is, ϕ is well defined. Because ag_0^{-1} assumes all values in G as a varies through G, we see that ϕ is onto L_K. If $\phi(aH) = \phi(bH)$, then $(aH)g_0^{-1} = ag_0^{-1}K = bg_0^{-1}K = (bH)g_0^{-1}$, so $aH = bH$ and ϕ is one to one. To show ϕ is an isomorphism of G-sets, it only remains to show that $\phi(g(aH)) = g\phi(aH)$ for all $g \in G$ and $aH \in L_H$. But $\phi(g(aH)) = \phi((ga)H) = (ga)g_0^{-1}K = g(ag_0^{-1}K) = g\phi(aH)$, and we are done.

Conversely, suppose that $\phi : L_H \to Ł_K$ is an isomorphism of the G-set of left cosets of H onto the G-set of left cosets of K. Because ϕ is an onto map, there exists $g_0 \in G$ such that $\phi(g_0H) = K$. Because ϕ commutes with the action of G, we have $(g_0hg_0^{-1})K = (g_0hg_0^{-1})\phi(g_0H) = \phi(g_0hg_0^{-1}g_0H) = \phi(g_0H) = K$, so $g_0hg_0^{-1} \in K$ for all $h \in H$, that is, $g_0Hg_0^{-1} \subseteq K$. From $\phi(g_0H) = K$, we easily see that $\phi^{-1}(g_0^{-1}K) = H$, and an argument similar to the one just made then shows that $g_0^{-1}Kg_0 \subseteq H$. Thus $g_0Hg_0^{-1} = K$, that is, the subgroups are indeed conjugate.

18. There are three of them; call them X, Y, and $\mathbb{Z}_4$ corresponding to the three subgroups $\{0, 1, 2, 3\}$, $\{0, 2\}$ and $\{0\}$, respectively, of $\mathbb{Z}_4$, no two of which are conjugate. The tables for them are

	X	Y	
	a	a	b
0	a	a	b
1	a	b	a
2	a	a	b
3	a	b	a

and essentially the group table for $\mathbb{Z}_4$ itself corresponding to $\{0\}$. (Conceptually, entries in the body of that group table should be in braces, like $\{3\}$ to denote the coset, rather than the element.)